

Operating Systems basics, security features, and virtualisation

Slides adapted from: CS124 CalTech, CS45 Stanford University,

Mihai Ordean

Summary

- OS components, memory layout/MMU, processes, threads
- Sandboxing as a security feature
- Virtualisation etc.

Operating Systems

Operating systems are used in many different contexts, for fulfilling many different purposes e.g.,:

- Mainframe and server operating systems must maximize utilization of hardware
 - Operating system doesn't require a graphical user interface
 - Rather, must support very efficient handling of I/O, and possibly scheduling of many processes
- Personal computers must be easy to use, and responsive to user input
 - Maximizing hardware utilization is less important – responding to user interaction is top priority!
 - Much more code is devoted to making the computer easy to use
 - Important to provide a simplified, user-friendly user interface

Operating Systems

Mobile device / tablet OSes have several challenging, often conflicting constraints

- Must be responsive and user-friendly, like PC operating systems
- But, must also try to maximize battery life through careful hardware resource management

Smartphones, must support download, installation, execution, and uninstallation of wide range of applications

- But, basic device capabilities (e.g. voice calls, SMS) must also be rock-solid reliable
- Must support intermittent connectivity, particularly when programs are using that connectivity!

Operating Systems

- By far the most common kind of computer now is the embedded computer
 - In your microwave oven, your printer, your WiFi router, your DVD player, controlling your car engine, your point-and-shoot camera
- Embedded OSes tend to have very limited capabilities
 - Systems tend to support a specific, fixed set of tasks
 - Systems aren't designed to run arbitrary programs on them
- Can still include a variety of basic OS capabilities
 - Basic thread-management and scheduling support
 - Basic memory management capabilities
 - Support for software upgrades
 - Support for peripherals like flash cards, USB drives, networking, ...

Operating Systems

Real-time operating systems focus on completing tasks by a specific deadline

Most general-purpose operating systems provide soft real-time support, e.g. for media playback

- Not considered a system failure if the OS misses a deadline from time to time (e.g. your media playback just sounds choppy)

Some real-time OSes provide hard real-time guarantees

- If the OS misses a deadline, this is considered a fatal error!

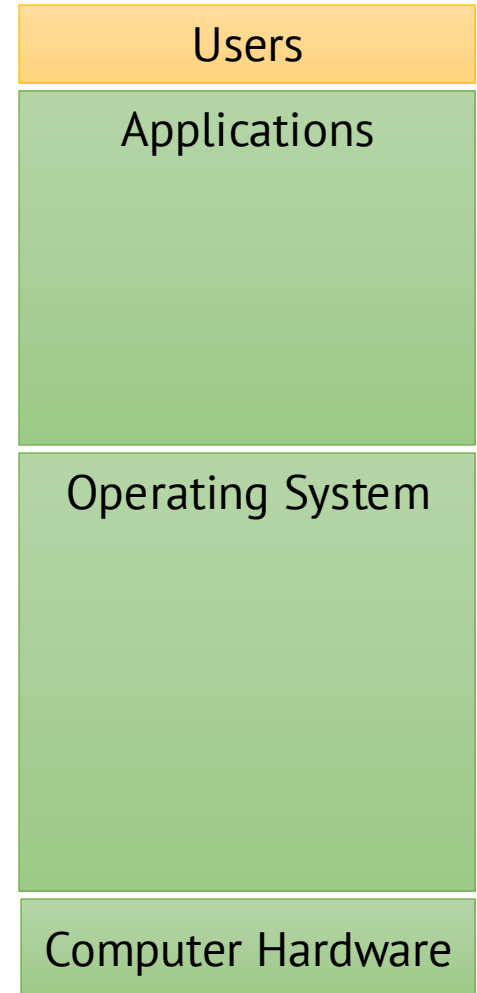
Example: a computer system for running an automobile manufacturing assembly line

- The OS receives inputs from sensors along the assembly line.
- If the OS doesn't satisfy guarantees for processing input data and controlling automated machinery, physical damage will occur
- If OS misses its timing deadlines: Failure! Halt the assembly line!

OS Components

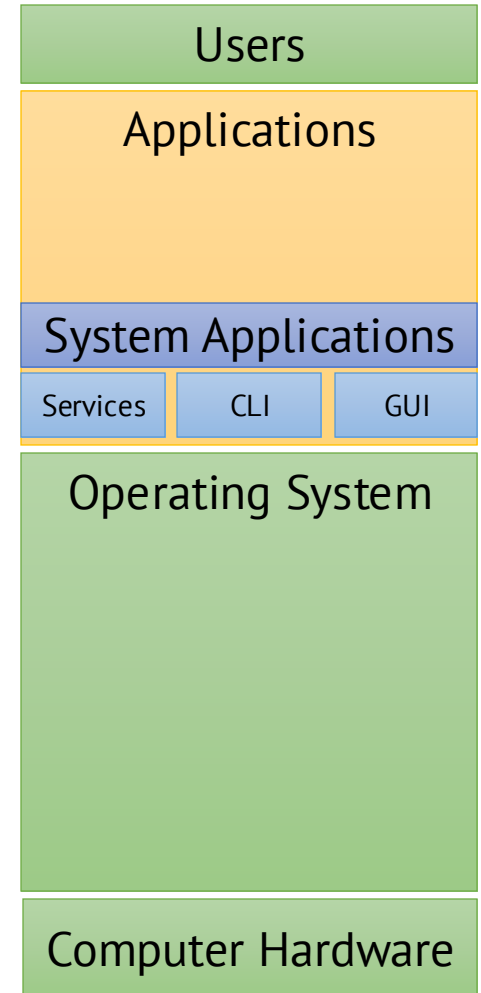
Common components of operating systems:

- Users:
 - Want to solve problems by using computer hardware
 - OS may support only one user at a time, or many concurrent users, depending on system requirements
 - Some systems usually have no users, so they have an extremely minimal UI
 - e.g. automobile engine computers



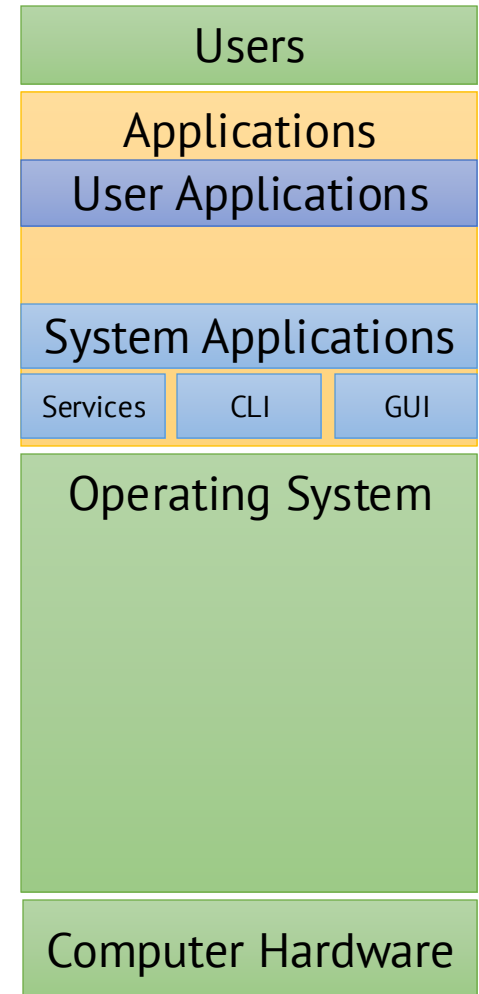
OS Components

- Common components of operating systems:
- Applications allow users to solve problems with the computer's resources
 - Applications rely on the OS to manage those resources
- Some applications are provided by the operating system
 - Services for providing and managing system resources
 - Command shells (e.g., bash, zsh, sh)
 - GUI programs (X-server, system config tools, etc.)



OS Components

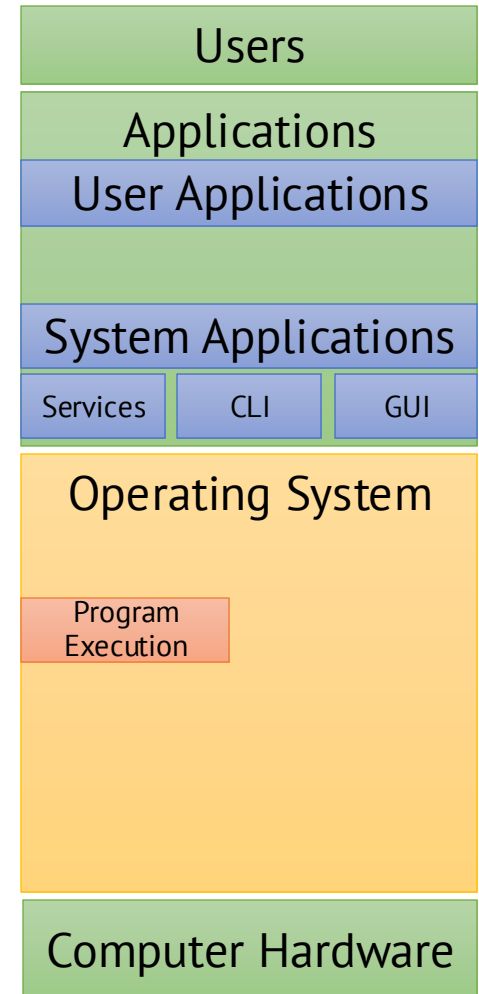
- Common components of operating systems:
- Applications allow users to solve problems with the computer's resources
 - Applications rely on the OS to manage those resources
- User applications are designed to solve specific problems
 - e.g. text editors, compilers, web servers
 - e.g. web browsers, word processors, spreadsheets



OS Components

Common components of operating systems:

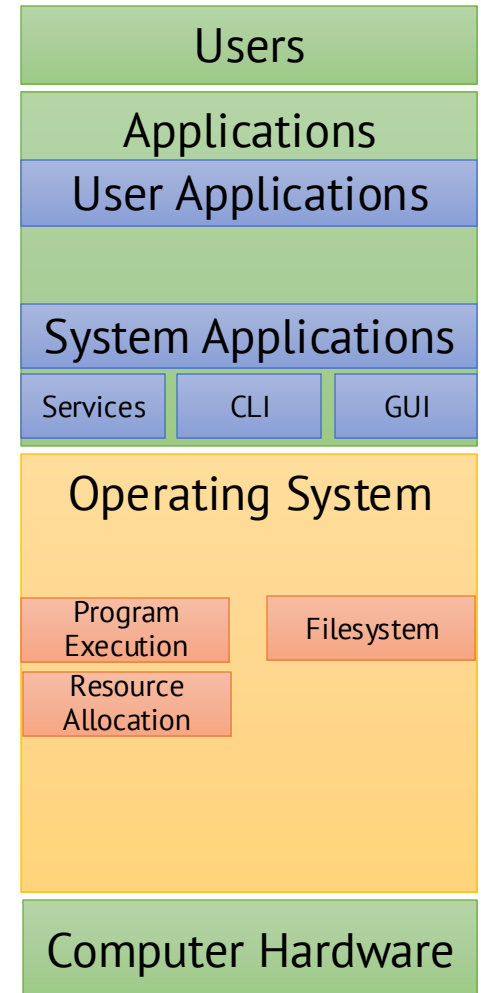
- The OS itself can provide many different facilities
 - Not every OS provides all of these facilities
- Most obvious facility: program execution
 - Load and run programs
 - Perform runtime linking of shared libraries
 - Handle program termination (possibly with errors!)
 - Pass along signals, etc.



OS Components

Common components of operating systems:

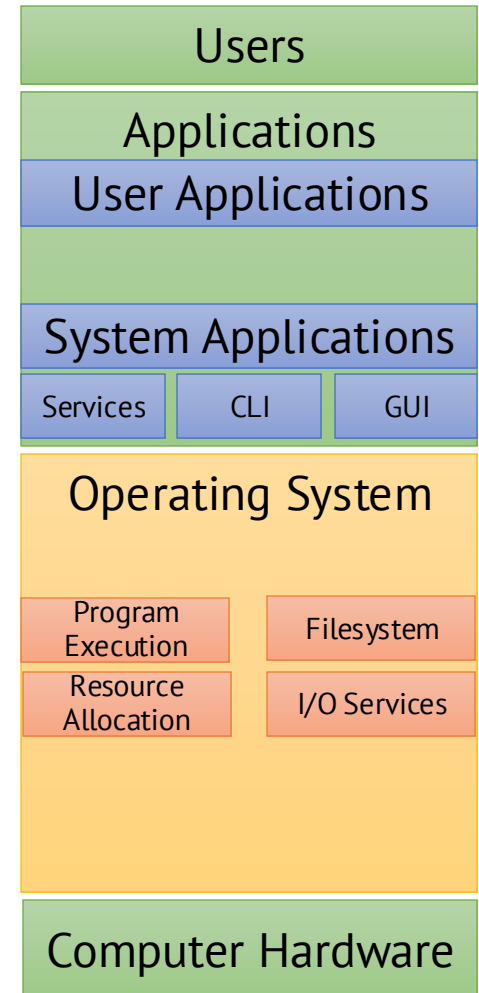
- Another obvious facility: resource allocation
- Resources to manage:
 - Processor(s) – especially if OS supports multitasking
 - Main memory
 - Filesystem/external storage
 - Other devices/peripherals
- Filesystems:
 - OS usually supports several different filesystems
 - May also require periodic maintenance



OS Components

Common components of operating systems:

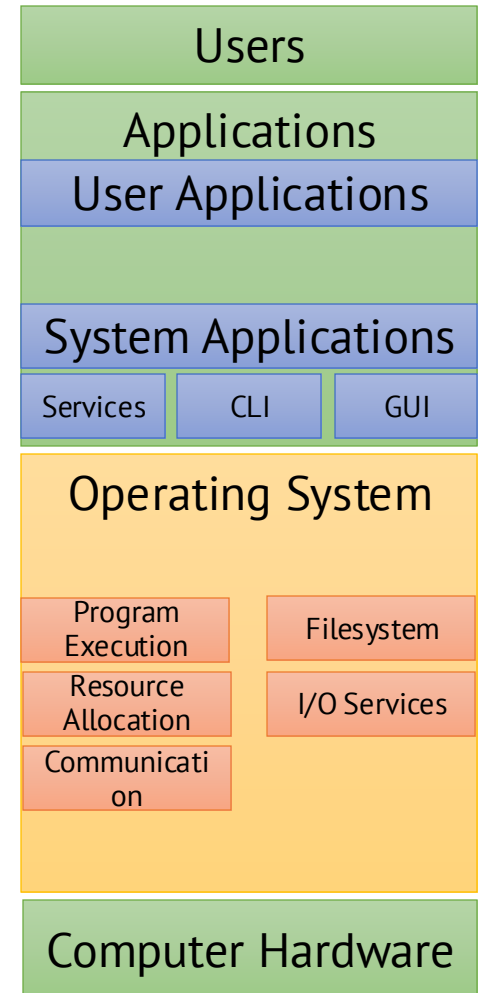
- Disks and other peripheral devices require specific interactions to function properly
 - I/O subsystem provides facilities to control computer hardware devices
 - Often interact via I/O ports
 - **Do not** want apps to do this!
- Usually modularized by using a device-driver abstraction
 - Present a clean abstraction for the rest of the OS to use
 - Encapsulate gory details of talking to hardware



OS Components

Common components of operating systems:

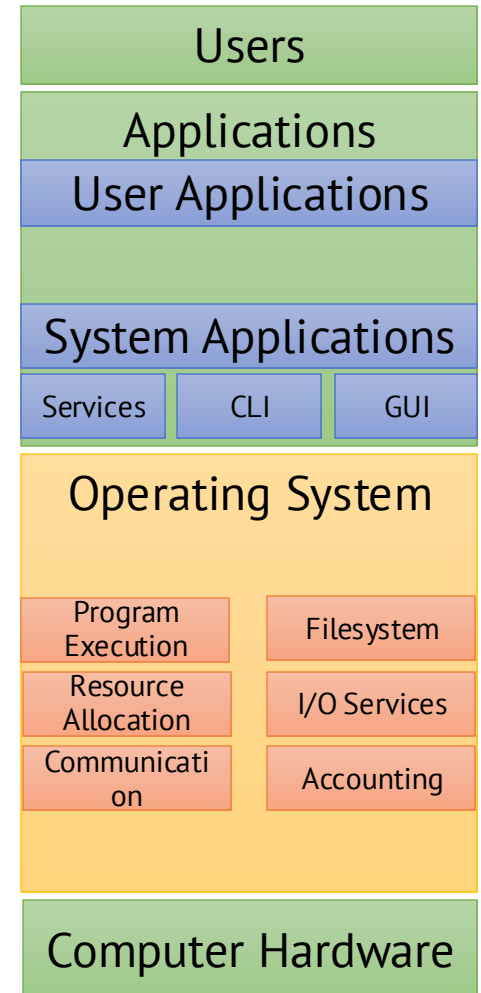
- OSs require internal and external methods of communication:
- Collaborating processes need to share information
 - Called **Inter-Process Communication (IPC)**
 - Many mechanisms: pipes, shared memory, message-passing, localhost, Unix sockets, d-bus, etc.
- Some processes need to communicate with other computer systems
 - Many kinds of networking



OS Components

Common components of operating systems:

- Some OSes record resource usage data:
 - Accounting facility
- Purpose: systems that enforce per-user restrictions based on CPU usage, storage, network, etc.
 - Very common to bill customers for storage and network use
 - Also, with hypervisors, very easy to bill per-VM for CPU use



OS Components

Common components of operating systems:

- OSes must handle various errors that occur
- Varies widely, depending on what the hardware can detect

Common errors:

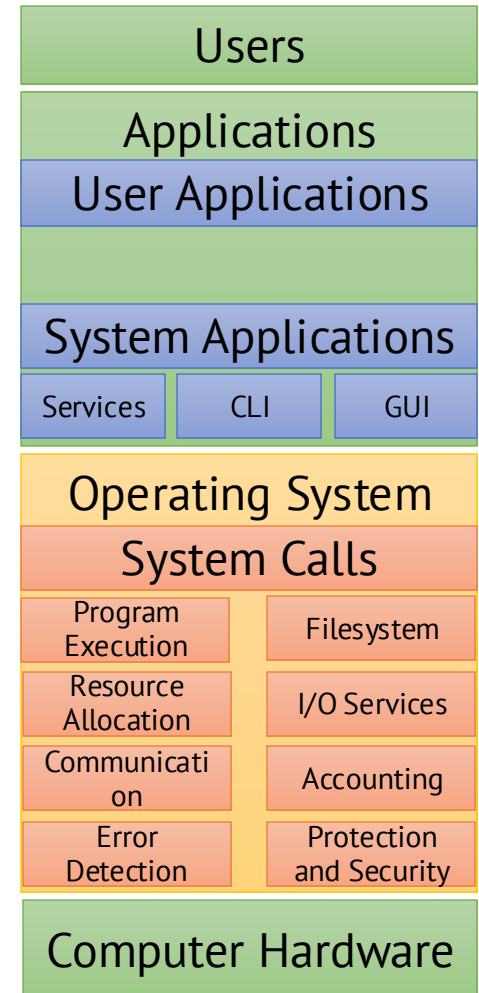
- Hard disk is full, or broken
- Filesystem is corrupt

Memory errors

- A program behaves in an invalid way
- Printer has no paper or ink

Less common errors:

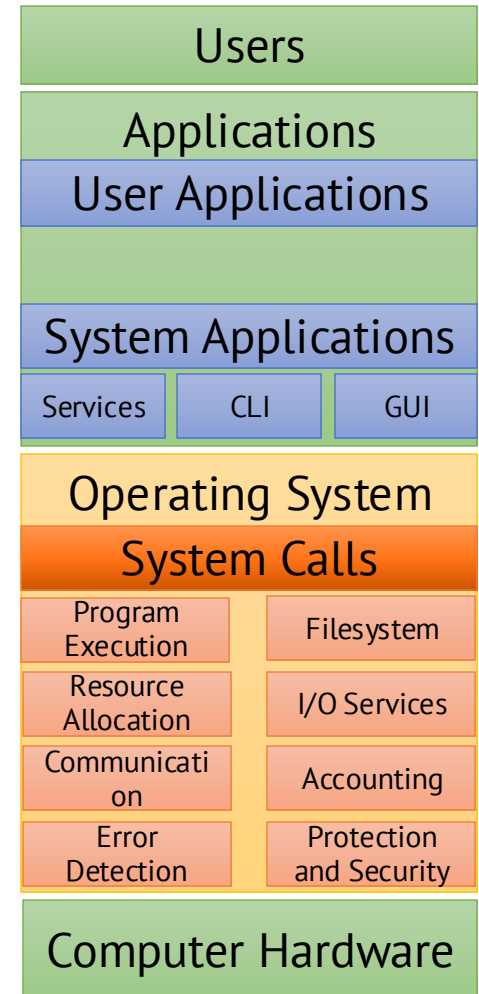
- Processor failure, etc.



OS Components

Common components of operating systems:

- OSes must prevent different kinds of abuses
- OS must be able to protect itself from malicious programs
- Applications are not allowed to directly access operating system code or data
 - Computer hardware must provide this capability!
- All application-interactions with OS are performed via **system calls**



OS Components

Common components of operating systems:

- Operating system must also protect processes from each other
 - A process should not be allowed to access another process' data, unless this is specifically allowed by the process,
 - i.e., the process has built in communication capability
- Again, this requires specific support from the computer hardware

